



BEYOND TRT

PREGNENOLONE, DHEA, THYROID,
ESTROGEN, HCG & FERTILITY

"Dave is the man I always strived to be before I even met him. He embodies so many good qualities that I value. But one of the key ones that led me to working with him was his transparency. He is so direct and honest in everything he does and that made me trust him with my health. I'm feeling so much better since working with Dave. I'll be a lifetime client with him that's for sure."

— Blake F

DAVE LEE

FOREWORD

A year ago, to the day, I wrote TRT 101.

The goal was to simply get my ideas down on paper.

To create a simple, referenceable guide where men all over the world could follow a template to understand a simplified, actionable approach to what I consider “best practice” testosterone replacement therapy.

In the year since, the book has been incredibly well received, and the reception well exceeded any expectations I had. This sequel, “Beyond TRT” which was promised in the pages of the first book, has been heavily requested and has been a pleasure to put together.

There are so many peripheral concepts that play an integral role in both the application and outcomes of testosterone replacement therapy.

This is a roadmap designed to guide you with all the actionable tools and information you will ever need to make the right start with TRT.

Believe me, I get it.

I was there once too.

And I have spent the recent years not just researching and discussing these topics with some of the best in the world, but also listening to and treating thousands of clients in not just TRT, but these peripheral or “troubleshooting” topics to get the best outcome for them.

By the time you finish this book, you will have the foundational knowledge equipped to deal with everything that goes beyond TRT.

I will cover the other hormones that can be deficient such as Pregnenolone, DHEA and Thyroid, as well as teach you how to avoid the pitfalls around fertility, HCG and estrogen.

Let's get started.

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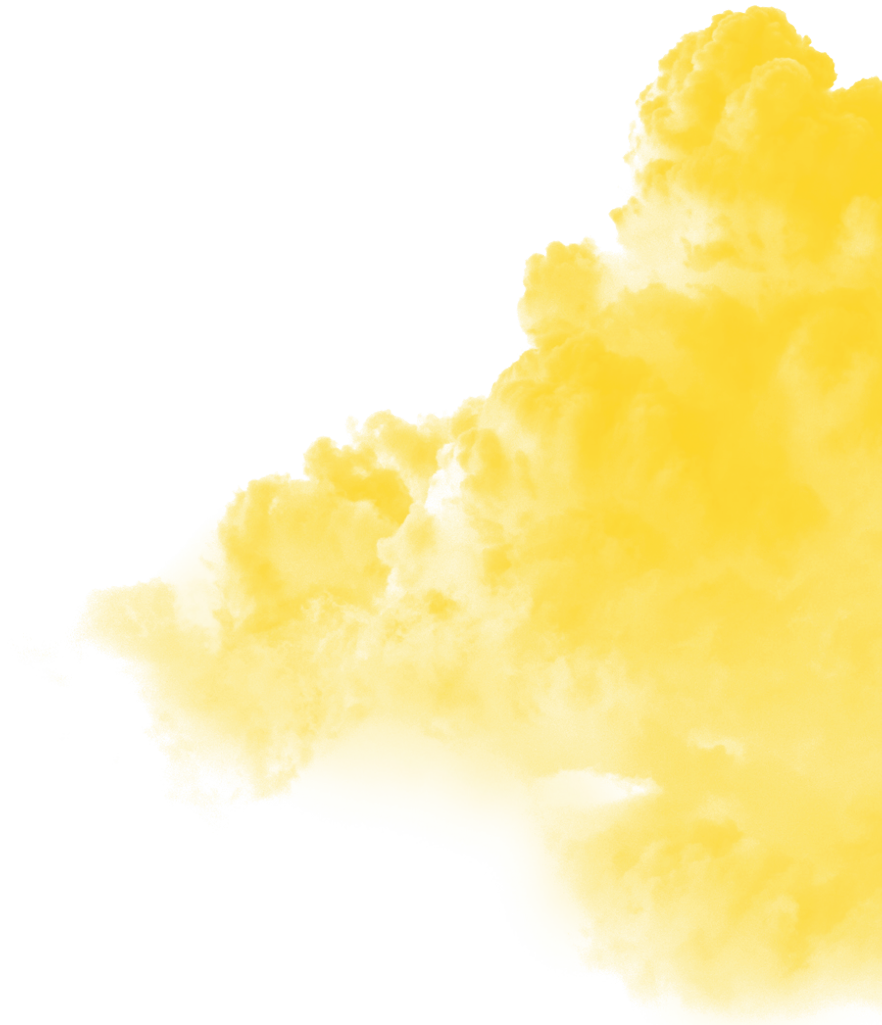
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CHAPTER I

INTRODUCTION - WHY
CAN'T IT BE AS SIMPLE AS
1, 2, 3?

Here's the sales pitch many guys get from their clinic, or under-trained MD, that gets them off on the wrong foot.

- ① Take this testosterone
- ② Get your blood work and we'll adjust the dose to get it right
- ③ You'll be dialled in, and that's all there is to it.

This SOUNDS fantastic.

If only it were that simple.

Unfortunately, many patients and practitioners alike believe one, simple fallacy: all of a man's problems can be solved with testosterone.

Tired, lethargic, depressed, anxious?

Up the dose.

Bloated, emotional, over stimulated or not sleeping?

Drop the dose.

Rinse, repeat.

"All of the solutions to your problems can be solved by simply tweaking the administration and dose of testosterone."

The human body simply is not that simple. Just like how all the issues with your car can't be fixed by tweaking or upgrading the engine.

There's a very simple saying I have when it comes to everything beyond TRT - ***where there's smoke, there's fire***. And while a good handful of men simply just need to fix their testosterone levels and everything works as promised, there are just as many men who need attention elsewhere, and these deficient areas can be complex, interconnected and multifactorial.

Contrary to this reductionist and overly simplified approach, testosterone is not the sole driving factor of a man's physical and mental health. It is a major component and has far reaching acute and chronic benefits, but if there is a deficiency elsewhere (or a "loose part", returning to the car analogy), no amount of testosterone is going to fix it.

In fact, throwing MORE testosterone at the equation might make it worse.

It can often be a hard pill to swallow for men who have started TRT to realise that interventional endocrinology is more complex than they were sold to begin with, and it is even more daunting to realise how much conflicting information (or lack there-of) there is surrounding these topics of health.

But this is the reality of the situation.

Here are some key areas to begin wrap your head around before we go on:

- 1 If you have a testosterone deficiency, you may have other hormonal deficiencies too. *Testosterone deficiency can occur in isolation in some, while hormone deficiencies may occur together in others.*
- 2 If the root cause of hypogonadism is secondary (*i.e. not related to failure of the testes known as primary hypogonadism*), the root cause of this dysfunction must be identified and either resolved or supported for TRT to work properly.

For example, if secondary hypogonadism is caused by obesity, *simply tripling the testosterone levels will not automatically resolve the obesity*, and there are likely other health consequences (including, but not limited to hormonal issues) still at play.

- 3 Not everyone responds in the same way to TRT, and while there are averages and trends, there is a large deviation away from the norm both in responses to doses, but also outcomes. *Your timeline, symptomatic improvement and overall outcomes may have less in common with the status quo than you would assume.*

- 4 Our biochemical individuality, the combination of our different genetics and years of living with different inputs and outputs as human beings, mean that comparing two men on TRT's journey is like comparing apples and oranges. *They're both fruit, but they're just as different as they are similar.*

For example, troubleshooting online by comparing your blood work, symptoms and progress to others may create more confusion than clarity. This often leads to frustration and confusion in patients, and re-enforces the need to work with a practitioner who understands the content laid out in the book, so that you don't have to do all the heavy lifting yourself.

If you're beginning TRT or are on the fence and are getting ready to pull the trigger - this is not a forewarning to scare you off.

However, it is forewarning to understand that hormone replacement therapy is something that requires proper medical supervision by an informed and competent provider, with whom you can have a conversation about the topics in this book, as well as the topics covered in TRT 101.

? 1.1 WHY ME?!

"My buddy started taking TRT and after 12 weeks he felt like Superman. How come it's not that simple for me?"

When troubleshooting TRT, this is one of the most common queries I have observed.

And the answer is very simple.

- ! **Testosterone replacement therapy will not solve any problem that was not caused by low testosterone.**

And believe me, not everyone's problems are caused solely by low testosterone, it's just a piece of the puzzle.

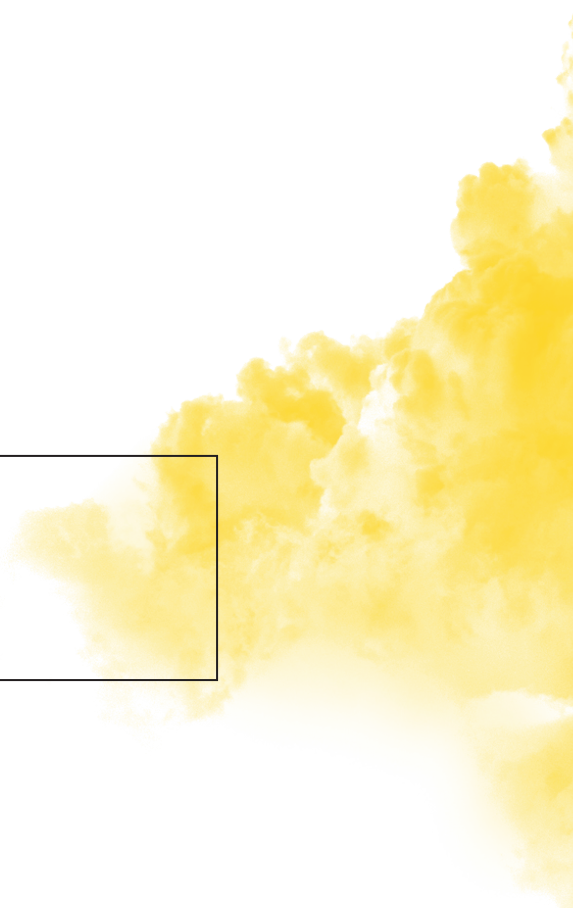
As you delve deeper into the chapters of this book and into each individual category, I'll be unpacking the root causes of why an individual may have a problem, and you'll notice a trend.

Many of these issues are caused by diet and lifestyle.

And while there may be interventions that can be made in regards to altering your HRT protocol, these catalysts may only be part of the solution.

Just like if we fix or upgrade your car, you still have to learn to be a better driver.

And nothing damages the human body like driving it badly over time.

A decorative yellow cloud-like graphic with a soft, painterly texture, located on the right side of the page, partially overlapping the text box.

CHAPTER II

HCG

2.1 INTRODUCTION

HCG is a controversial and difficult medication to work with when it comes to TRT.

Despite what may be written online, beyond the applications for fertility (outside of TRT) the effects of HCG on men are not well studied, and not well understood.

Although HCG has been studied in high doses for long periods (studies of up to 1500IU 3x weekly for 10 years have been conducted) the outcomes beyond direct fertility parameters were not assessed.

In reality, when it comes to how HCG is typically prescribed with TRT, we don't know as much as we think we do when it comes to what this is doing systemically in the long term.

What complicates this further is that HCG is an easy upsell to make on top of TRT, and along with AIs (they're up next) they are the most "upsold" ancillary with a TRT prescription.

I liken the HCG and AI combo to the "would you like fries and a coke with that" at McDonalds.



Quick fun fact: The majority of McDonald's profit margin is in the fries and drink, not on the burger itself. While TRT is profitable for clinics to prescribe, it is much more profitable, for example, to have 100 men on 3 medications, than 300 men on 1 medication, as the latter involves 3x the amount of patient management. Just like all advertisements and marketing content, always take the advice of someone who is selling you something with a grain of salt.



Note: Not all doctors run clinics, and not all clinics make profits from medications. It's important to understand the integrity of your provider's advice, and the best way to get an insight into this is to understand this business model.

HCG is often touted as a Luteinizing Hormone Mimetic i.e. it mimics the effect of a hormone known as “luteinizing hormone” (LH) on the luteinizing hormone receptor and accordingly, we can simply replace the body’s luteinizing hormone with HCG while on TRT to keep this system “intact”.

This sounds great on paper, but is a misunderstanding and oversimplification of HCG, and these nuances may explain the high incidences of side effects in practice.

***Background:** When a man goes on testosterone replacement therapy or uses any form of exogenous (external) androgens, his body will sense this and reduce production of his own endogenous (internal) androgens. This is something called a “negative feedback loop”. Similarly, if you have a glass of water, the body will sense when the glass is empty and pour more water, and sense when the glass is full and pour less. This is how the body maintains “homeostasis” which is a system of balance, and functions like a thermostat to keep things regulated. When the androgen receptors are stimulated by external testosterone, the body stops pouring the water into the glass as it knows the glass is full, even if it isn’t filling up the glass itself. This reduces the signal from the pituitary gland to the testicles. As a result, Luteinizing Hormone and Follicle Stimulating hormone are “shut down” to zero. This causes the testicles to become dormant and non-responsive, as the body has replaced its internal testosterone production with an external testosterone source. Thus, testosterone **replacement** therapy.*

The primary issue here, is the term “luteinising hormone mimetic” and how this is applied.

The term mimetic is defined on dictionary.com as: “characterised by, exhibiting, or of the nature of imitation or mimicry”.

By this definition, HCG is not a luteinising hormone mimetic.

This, in my opinion, based on my research and observations, is why HCG is a “curveball”.

In the body, we have receptors that can be activated by multiple targets. For example, anabolic steroids will all bind with the androgen receptor, but they exert different effects from compound to compound, which is what makes them different.

DHEA binds to both the androgen and oestrogen receptors to exert its effects, but it does not exert the same effect as testosterone, or any of the bioidentical oestrogens.

LSD does not produce the same effects as serotonin when it binds to the serotonin receptor (trust me on that one).

These are just a small handful of examples to illustrate the simple point: not all agonists exert the same effects on the receptors to which they bind.

This is a fairly well understood concept in human biology, but in TRT the difference between luteinizing hormone and HCG is lost, as it is determined that they both activate the same receptor, *and that's that*.

However, this leaves two key questions unanswered:

- 1 Why does HCG cause side effects in men that were not caused by their own bioidentical luteinizing hormone? Why is it that some men do not tolerate HCG if it simply “mimics” a bioidentical hormone?
- 2 Why would a pregnant female produce HCG during pregnancy if it has the same effect as luteinising hormone? Why wouldn't she just make more of that?

Let's backtrack a little...

? What is HCG?

HCG, Human chorionic gonadotropin is a hormone produced by women when they fall pregnant, and is a hormone that is critical for many of the physiological outcomes of the first trimester.

It is a hormone that doubles every 2-2.5 days in a healthy pregnancy, and is the hormone that a woman is testing for when she urinates on a pregnancy test. If it's detected - she's pregnant.

HCG stimulates the luteinising hormone receptors in both men and women.

In fact, luteinising hormone receptors are fully known as luteinising hormones/HCG receptors, due to the critical role HCG plays on these receptors during pregnancy.

Men do produce HCG, but only minute and fractional amounts, leagues below the amount that we are using for TRT.

HCG has been fairly well studied in men for fertility, as high doses (often 1500IU up to 5000IU given in 3 doses per week) will stimulate both a man's LH and FSH receptors, increasing sperm production and can be an intervention to increase a man's fertility to the point where he can viably conceive.

? So, what does this have to do with TRT?

Because HCG can activate LH receptors, and to a lesser degree, the FSH receptors, it can be used to "replace" the luteinizing hormone that is shut down due to the presence of external androgens, and thus keep this cascade and testicular hormone production functioning.

This all sounds great. *There's just one small problem.*

There is a decent subset of men on TRT who simply do not tolerate HCG.

2.2 HCG MYTH BUSTING

MYTH #: "All men need HCG when on TRT."

On paper, it can absolutely be argued that mechanistically, HCG could be beneficial for all men on TRT.

However, in reality, not enough is known about HCG to ascertain whether it is more or less optimal for a man to be on TRT long term.

In my experience, around a third of men feel great when adding HCG to TRT. A third feel nothing at all (outside of larger testicles) and a third get intolerable, debilitating side effects from it.

For this third, it doesn't matter what mechanistic and hypothetical benefits HCG carries, if it makes them feel awful and TRT makes them feel great, it's common sense.

They're better off without it.

MYTH #: "TRT without HCG will destroy a man's testicles."

While the testicles are stimulated in utero, from birth until the beginning of puberty, the testicles do not receive any signal from luteinizing hormone or follicle stimulating hormone. It is the beginning of this signaling, amongst other things, when a child becomes of age, that begins the process of puberty.

Based on this logic, it be an assumptive leap that TRT administered without HCG will cause the testicles to undergo "irreversible and irreparable damage", a myth that is pervasively touted by many TRT clinics that sell HCG based on the most effective marketing tactic known to man.

Create fear around a problem you didn't know you had, and then sell the solution.

TRT in the absence of HCG will cause the testicles to become around 30% smaller or lighter than prior to TRT, and this will vary based on the individual. It is simply not known whether the testicles will become more or less responsive to HCG or other LH receptor agonists without prolonged lack of a signal.

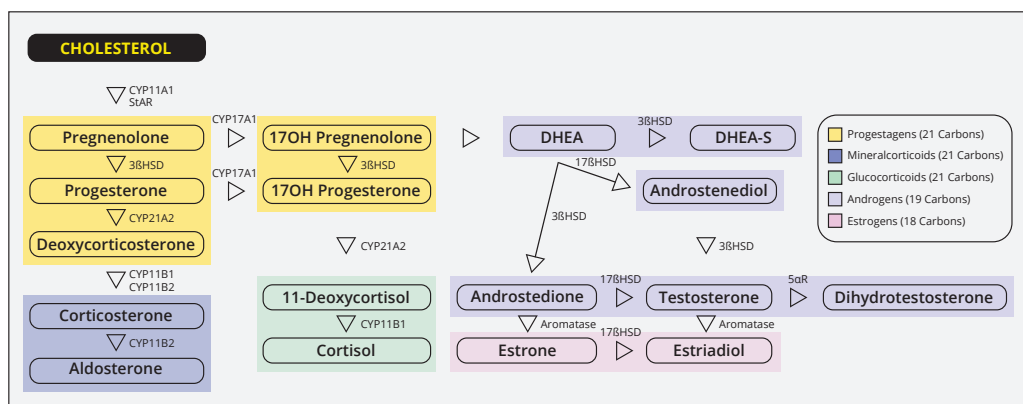
However, one thing is for certain, is that a small period of utilising testosterone monotherapy will not “destroy” your testicles. This is a baseless claim. See my recommendations at the end of this chapter for how to navigate this predicament.



MYTH #: “HCG increases levels and/or prevents shut down of “upstream” hormones Pregnenolone and DHEA.”

This is another extremely pervasive myth that does not have the evidence in literature, nor the outcomes in practice, to back it up. This is a phrase that is often parroted by patients, doctors and clinics alike, and stems back to a hypothesis made by the late Dr John Crisler (RIP) not long before he passed.

Unfortunately, this hypothesis has been deemed fact by many enthusiasts in the field, and while pregnenolone and DHEA will be discussed at length in later chapters, there is an integral point to understand here:



This diagram shows the “steroidogenic hormone cascade”. It can be seen that all hormones begin with the conversion of cholesterol to pregnenolone, and then are created via a series of enzymatic processes from one another.

However, this has been interpreted far too literally.

While this diagram illustrates the pathways, it should not be interpreted as a literal river, or as water pipes with hormones “flowing down a stream”.

The human body simply does not work like this, and this drastic oversimplification has created an enormous amount of confusion on the topic.

Although LH/HCG is one of the signaling molecules to produce the enzyme CYP11A1 that converts cholesterol to pregnenolone, it is not the only signal.

These hormones are not only made in the testicles, they are also made in the adrenal glands. In fact, DHEA is primarily made in the adrenal glands, while pregnenolone is also produced in the adrenals and brain, and these pathways are not impacted by luteinizing hormone.

If they were, we would see DHEA and Pregnenolone levels fall to zero (or close to) in patients on testosterone monotherapy, and in practice, this simply is not the case.

In practice, as someone who has monitored thousands of blood tests for testosterone, testosterone + HCG, as well as pregnenolone and DHEA deficiency, I have never seen HCG restore deficient levels of DHEA or pregnenolone.

In fact, it often doesn't move it much at all.

While men on TRT may become (or are already) deficient in DHEA and Pregnenolone (discussed in later chapters), the use of HCG does not impact this outcome to a significant degree.

This does not mean HCG does not have benefits, as it does impact intratesticular hormone production, but this pathway is not one of the beneficial or relevant roles of HCG.



MYTH #: "HCG prevents TRT causing infertility."

Fertility and TRT is yet another extremely controversial and sensitive issue.

And rightly so.

However, many men are sold the concept of HCG as an insurance policy to prevent, or even a bulletproof vest to protect against infertility, and this is far from the truth.

It is worth defining from the get-go, that TRT does not cause infertility in all men. If it did, testosterone would be an extremely popular and widely used contraceptive.

It's not.

Quick side joke: What do you call the body-builder who thought his steroid cycle would act as birth control? Dad.

However, if TRT was to hypothetically cause infertility in all men...

(Again, it doesn't. I get into this in more depth later)

HCG is not an effective prophylaxis (preventative) against this.

This is for a number of reasons, but as we identified previously, HCG is primarily a luteinizing hormone receptor agonist, NOT a follicle stimulating hormone receptor agonist.

HCG will maximally stimulate intratesticular testosterone production at around a dose of 250IU-500IU per shot, depending on the study.

However, for fertility purposes, HCG is often used at significantly higher doses: 1500-5000IU per shot. In THESE doses, it effectively, to a degree, stimulates the follicle stimulating hormone receptor.

But in the doses that are well tolerated for an adjunct to TRT (500-1500IU per week in divided doses, typically) it may do this a little, but if it does, it's not much.

This is why time and time again, clients end up in my office, and in many practitioners' offices, who are infertile while on the TRT + HCG protocol that they were sold as a prophylaxis for this very problem.

In reality, HCG in high doses is a moderately effective fertility agent in men. However, these doses, in addition to TRT, are not well tolerated or advised in the long term.

While the doses of HCG that are typically used with TRT do a great job in stimulating the testicles and keeping them full and active, it works marginally, at best to prevent infertility.

In practice, TRT does NOT make all men infertile, but it does absolutely compromise the fertility of some.

If your fertility is compromised on TRT via a sperm test, a maintenance dose of HCG will yield little effect on your 3-6 month follow up sperm count.

2.3 HCG VERDICT

HCG is a curve ball for patients and practitioners alike. However, as discussed and identified, it is far from a “must-have” addition for all TRT patients.

As outlined in TRT 101, I recommend not beginning your TRT with HCG, as there is a significant chance that you will not respond to it favourably. It can be difficult to identify this if you do, as you can have one thing making you feel better and one thing making you feel worse, and it's hard to identify the best course of action when you experience side effects.

Instead, I recommend that if you, or your practitioner want you to trial HCG, it is done after testosterone levels have been adjusted, and hypogonadal symptoms have begun to improve. For most, this will be within the first 6-12 months of therapy.

If stopping testicular stimulation for 6-12 months will destroy your testicles, they wouldn't have survived the decade (and then some) with no stimulation prior to puberty.

There are some men who will receive more testicular atrophy than others, and this may cause discomfort during intercourse or when exercising, and for these men, HCG will be more of a priority to trial. However, this tension and retraction of the testicles into the abdomen can often be resolved with pelvic floor therapy and improving mobility in the hips and abductors.



Hint: Always look for what you can “do”, before you look for what you can “take”.

The bottom line on HCG is that while there can be mechanistic arguments for why it is, or isn't needed, it is only your experience that will show if you react positively or negatively to its inclusion with your TRT protocol.

I have seen guys report drastically improved mood, energy, libido and sexual function, and say it's like “the lights come on” mentally for them.

Meanwhile, I have had other men state that HCG causes debilitating anxiety, sweating, insomnia and mood disturbances that cannot be rectified with dose reductions or aromatase inhibitors, all while producing little to no changes in their serum hormone levels.

If HCG is to be used with TRT, due to its short half-life of 24-36 hours, and propensity to cause side effects, I prefer it to be dosed at 100-150IU daily, or 250IU 3x weekly/every other day. If a twice weekly protocol is followed then the dose, in my opinion should be 250IU-500IU 2x weekly, but I think 3x weekly minimum is best.

Generally, if HCG is going to cause a negative response, it will occur within hours or days. If HCG doesn't make you "feel" anything within the first week, this is a good sign, as testicles often take 6-8 weeks to restore in size.

Keep in mind that HCG is not a libido or sexual function panacea although it may be touted as one in online forums.

If libido is low due to chronic stress, obesity, other hormonal issues or relationship/lifestyle problems, HCG is not going to cancel any of these out. In my experience, the men who get negative side effects from HCG get a reduction in sexual desire and performance on it, so it truly can have different outcomes from person to person.

To date, neither me, nor anyone I have spoken to has been able to identify the common variables in men who respond positively, and men who respond negatively to HCG. That for now, remains a mystery.

I don't find it surprising that a decent cohort of men produce an undesirable reaction to injecting a physiologically significant amount of female pregnancy hormone, regardless of their intentions. It could very well be an immune system response that is prominent in some rather than others, but to me, I would find it surprising if 100% men optimally tolerated augmenting their physiology with a female pregnancy hormone, that was never intended to be present in these quantities in male biology.



CHAPTER III

AROMATASE INHIBITORS

3.1 AROMATASE INHIBITORS INTRODUCTION

Aromatase inhibitors, or AIs for short, are by far the most disagreed upon area of TRT.

This has progressed in the online TRT space to the point that the arguments often become hostile and degenerate into non-scientific discussions, and the “sides” of each debate have become cult-like in nature.

As outlined in TRT 101, I am of the stance that aromatase inhibitors do not have a place in optimal male health, for the reasons I will discuss here. However, I do believe that the men complaining of the “estrogenic symptoms” they are seeking to remedy do have **very real problems** which need addressing and that “estrogens” play a partial role in this pathology, but there is a different way to go about things.

And a lot more that needs to be unpacked.

To begin, I will lay out a couple of quick facts:

- 1 Every study to date showing the positive outcomes of testosterone replacement therapy has not inhibited aromatase.
- 2 There are no long-term studies on the use of aromatase inhibitors in men, especially when used in addition to testosterone replacement therapy.

This means that aromatase inhibitor use in men is the opposite of evidence-based medicine.

- ? Healthy young men with optimal testosterone levels do not need aromatase inhibitors, so if TRT is being used to replicate the levels of testosterone present in a healthy man, why would it cause estrogenic side effects?

The answer is common sense.

It's either the protocol does not represent the levels of a healthy man,
or
we are not talking about a healthy man.

? So, why is the idea of NOT using an aromatase inhibitor controversial?

Aromatase inhibition, the inhibition of the enzyme “aromatase” (CYP191A) that converts testosterone to estradiol via the process called “aromatisation” has been synonymous with testosterone use for a long time, due to its cross over from body building, as well as the treatment of side effects when testosterone replacement therapy is improperly administered.

If you've read TRT 101, you will know that testosterone cypionate or enanthate, the most common protocol for TRT, needs to be administered more than once a fortnight, or once a week, as it is commonly prescribed.

This is because these large, bolus doses create a substantial “peak” and “trough” inducing a rollercoaster-like effect, and does not represent natural male physiology.

Twice weekly, or more frequent injections such as 3x weekly, every other day (EOD) or even daily injections, are a must when utilising testosterone replacement therapy for the majority of patients.

And while some men may “get away” with less frequent injections and do just fine, this is the FIRST thing to look at when fine tuning or trouble shooting a protocol.

But injection frequency does not solve all problems.
It only solves the problems relating to injection frequency.

In my experience, there are a multitude of factors why a man may experience side effects on TRT, but there are some issues that arise more often than others.

These issues are as follows:

1 OBESITY

Obesity is one of the biggest health problems in the world today, and obesity plays into a vicious cycle when it comes to testosterone production.

Obesity suppresses testosterone production.

and

Low testosterone inhibits fat burning and lowers energy metabolism.

This “chicken or the egg” is one issue that practitioners and patients alike must consider.

? “Am I fat because I have low testosterone?”

or

? “do I have low testosterone because I’m fat?”

In my experience, it’s a bit from column A and a bit from column B for most.

What must be considered when initiating testosterone in an overweight individual, is that if the body is actively suppressing testosterone production, and you inject high levels of testosterone it will cause side effects. **This is because you are not working with your natural physiology, you are working against it.**

And it is no surprise that the outcome will be affected adversely.

We know that adipose tissue increases aromatase production. But the solution here is not to simply inhibit aromatase, as estrogen is vital for lipolysis (fat burning) as well as having an anti-inflammatory effect caused by the excess body fat.

The solution is to lose the fat.

If a pharmaceutical intervention is to be made to manage the excess aromatase production and estrogenic side effects from excess body fat, it makes far more sense to use a medication that has been scientifically shown to do just that - Metformin. I recommend discussing using 500mg 2x daily, up to 1000mg 2x daily, with your prescribing doctor.

This approach is working further “upstream” in the body. Rather than trying to suppress the end outcome issue, we are able to look further up the cascade of events and address not just the problem itself, but work towards a solution.

This is improving the insulin sensitivity of an obese individual that not only reduces aromatase, but also helps the individual lose body fat so that they don’t need a pharmaceutical intervention long term.

Another approach I have used successfully with my clients is to use a more modest dose of TRT with daily injections or topical cream, and then titrate the dose up, as the body fat is reduced. This makes the most “biological sense” as it is working with the body to prevent the need for a pharmaceutical intervention to be made in the first place.

Diet and exercise MUST be the first priority when it comes to reducing body fat. For more information about reducing body fat on TRT, or before starting TRT, check out Ali Gilbert’s GPP for TRT program, as well as our combined Mind and Muscle Masterclass.



More information on: <https://ali-gilbert.com>

2 FATTY LIVER DISEASE

Fatty Liver Disease, whether alcoholic or non-alcoholic (AFLD or NAFLD, respectively) is one of the most commonly overlooked and dismissed diagnoses from general practitioners I have ever seen.

The presentation of fatty liver disease is often without acute symptoms, but there are symptoms of malaise, low energy, varying degrees of insulin

resistance and hormone dysfunction. It is often, but not always alerted by an elevation in “bilirubin” in blood work, as well as elevations in AST/ALT depending on the severity.

If you have elevations in the marker “bilirubin” you must have a liver ultrasound to check for fatty liver disease.

Especially if you have elevated markers for “direct” bilirubin, but this marker is often not tested.



Note: Not everyone with elevated bilirubin has AFLD/NAFLD, and not every case of AFLD/NAFLD will present with elevated bilirubin above 20. Gilbert's Syndrome will cause an elevated total bilirubin, often above 20, but will not elevate direct bilirubin.

Although your doctor and radiologist may dismiss it as a “don’t worry about it” scenario, it is important to understand that fatty liver disease will hamper your weight loss efforts and hormonal function and, over the course of your life will progress into a chronic disease state that may end in more severe liver disease such as cirrhosis or liver cancer.

It is easily treatable and reversible, and the myth that it is only caused by alcohol and obesity is not the reality.

Studies have shown that fatty liver disease can often pre-date obesity and type 2 diabetes, rather than it being the outcome of these states.

I often see clients who have tried for years to reduce body fat and improve body composition, even with TRT, and their efforts are in vain until they treat their fatty liver disease.

Disruptions in liver function, whether acutely from alcohol or chronically from liver disease is a silent and often neglected cause of side effects on TRT. It is why I always run liver panels when trouble shooting for my clients, and it is very common for their side effects to resolve when fixing their liver.

For a more in-depth breakdown of liver disease and supporting your liver naturally, I suggest watching my free liver masterclass on my friend Danny Bossa's YouTube channel.



Liver Masterclass - <https://www.youtube.com/watch?v=v7PAtinpFel>

It is not clear in the literature why NAFLD can occur prior to obesity and insulin resistance. My hypothesis is that NAFLD can occur due to a genetically higher requirement for, and subsequent deficiency in the key nutrient for liver health, choline. This nutrient can be sourced in the diet from egg yolks and liver, as well as in lower amounts from red meat. Egg yolks are the greatest source of choline.

Fatty liver disease can easily be treated with 1-2 grams of each choline and inositol, twice daily, for 8-12 weeks depending on the severity. I recommend a gram of each twice per day for 8 weeks for mild cases, and 1.5-2 grams for 10-12 weeks for more severe cases. It is important for ZERO alcohol to be consumed over the treatment period.



Note: It is integral to discuss any treatments for NAFLD/AFLD, or anything else mentioned in this book, with your prescribing doctor.

I also like to include glutathione, at either 500mg-1g daily of an oral reduced form daily, or 200mg of injectable glutathione twice per week.

If the gallbladder is obstructed by the liver fat on the ultrasound, I include 500mg of TUDCA daily during the treatment to support bile function.

Once the fatty liver has been resolved, I like to use a botanical liver support prophylaxis for 6 months containing ingredients such as Milk Thistle, NAC, Phosphatidylcholine, ALA and other liver support nutrients, to prevent recurrence and support general liver health. My preference is 1 serving daily of Now Foods Liver Refresh, 2 capsules daily of Paradise Herbs Milk Thistle, or 1 capsule daily of LVLUPHEALTH's Liver Complex (Lee10 for a discount from LVLUPHEALTH).

3 HYPOTHYROIDISM

A deep dive into hypothyroidism, its causes and treatments are elaborated later in this book. For now, I will be focusing only on its relationship to estrogen and TRT side effects.

? Did you know, one of the main regulators for aromatase production is T3?

The thyroid is one of the most critical glands for health and wellbeing in the human body. So, it is no wonder that when its function is damaged or dysregulated, things go astray, especially when exogenous hormones are introduced.

When an individual presents with the common side effects from TRT and points the finger at their serum estrogen level, it is instead prudent for the practitioner to perform an examination of thyroid function.

There is a strong relationship between obesity, liver function and hypothyroidism.

For example, a sluggish liver will increase insulin resistance and reduce conversion of T4 to T3, contributing to both obesity and hypothyroidism.

Obesity will have another vicious cycle relationship as mentioned previously with the thyroid gland, as while the correlation between obesity and hypothyroidism is often attributed to thyroid as the root cause

There is sufficient evidence to suggest that the increase in systemic inflammation *from* obesity is also capable of dysregulating thyroid function, increasing reverse T3, slowing thyroid function and making weight loss that much more difficult.

And, to top it off, hypothyroidism can both cause, and be caused by...

Drum roll...

Fatty liver disease.

This triad of metabolic dysfunction and chronic disease is one of the most overlooked and neglected contributors towards becoming locked in a stubborn state of dysfunction.

Aromatase inhibitors are not the cure, they are simply band-aiding the problem.

And, unfortunately many men are told their body fat is “fine”, their liver is “fine” and their thyroid is “within range”, when in reality, they have a problem with either, or all of the above, they just haven’t found a practitioner who can diagnose and treat these issues properly.

3.2 AROMATASE INHIBITORS MYTH BUSTING



MYTH #: “There are genetic over-aromatisers.”

One of the most common scape-goats for avoiding the reality of diet and lifestyle consequences is blaming your genetics.

While some people will be genetically disposed to autoimmune thyroid disease (or thyroid abnormalities across the board), developing fatty liver and a plethora of variance in other metabolic functions, I have not been able to find any singular nucleotide polymorphisms (SNPs) that explains why one guy will need an aromatase inhibitor and another won’t.

There are hundreds of SNPs relating to estrogen, across the various receptors, metabolism and the function of aromatase itself, and there will be varying polymorphisms and mutations across the board that account for our individual biochemistry.

If there are epigenetic shifts that cause a man to over aromatise due to a prolonged androgen-deprived state, these shifts would be reversed following chronic exposure to optimal androgen levels. This could account for another

reason why TRT takes time to work.

When I have spoken to men who claim that they are “genetic over-aromatisers” there has been something off in their diet and lifestyle that they do not take accountability for, or there is a pathological state not diagnosed.

Or, they are simply overweight.



MYTH #: “Being leaner than 15% body fat is not good for your health and is only for vanity reasons.”

When I came into this space from a neuroscience background, I would have never understood the importance of being leaner for better health.

If you think having a six pack is only good for taking your shirt off on the beach, think again.

You must get *ridiculously* lean as a man for it to become detrimental to your health, and excess body fat beyond 15% is causative for insulin resistance.

The more body fat you have, the more inflammation and the more metabolic dysfunction.

Many men could improve their health and response to TRT by reducing their body fat to 10%, or into the high single digits. If you're having side effects on TRT and don't like how you look naked - you know what you need to do.



MYTH #: “If you get side effects from anastrozole, you’ll do better with exemestane, and vice versa. You need to dial in the dose to find your estrogen “sweet spot”.”

*From my observations, aromatase inhibitors, when dosed conservatively, can improve negative side effect symptoms from TRT **in the short term.***

However, they often lead to a suite of new problems and side effects in the long term, that cannot be remedied by finding the “sweet spot” or swapping the class of the drug.

While these drugs differ drastically in their mechanism of action (I would choose to use a daily micro dose of exemestane if I had a gun to my head and had to use one of them) their outcomes are largely the same.

Reducing your aromatase activity via a drug is not going to lead to an optimal state of male physiology.

In fact, using these medications to band-aid the negative signal from your body may then result in you neglecting to resolve the root cause problem, and will lead to more negative outcomes in the long term.

Following the above approach, I have not had any clients who have “needed” aromatase inhibitors to feel their best on TRT.

I have seen many men try to take AIs, and discontinue them after a few weeks or a few months due to side effects, even when taken in doses to only reduce their serum estrogen by 20-30%, to the number they thought was the “goldilocks” zone.

Remember, estrogen is a paracrine hormone in men, meaning that estrogen is made locally in the cell, rather than being transported around the body for use in the blood like testosterone. *This means that the serum number is a “proxy” for total estrogen activity and it is an imperfect one.*

Rather than trying to fix the symptoms or the by-products of a dysfunctional state, if you are having side effects from TRT, it is vital that you have a comprehensive assessment from a knowledgeable practitioner who can diagnose and treat the root cause condition.

Healthy men with optimal testosterone levels don't need aromatase inhibitors, and a well-constructed TRT protocol mimics this physiology. You don't need one either.

3.3 AROMATASE INHIBITORS VERDICT

In my experience, there can be two hard pills to swallow when it comes to understanding aromatase inhibitors:

- 1 You can't blame your genetics for your problems
- 2 Testosterone takes time to turn you into the man you want to be

Rather than seeing “estrogenic side effects” as a problem that must be corrected with pharmaceuticals, it is more important to look at what can I “do”, or what do I need to “change”, rather than what can I “take”.

The other factor seldom discussed, is that testosterone takes time to androgenise your psychology and physiology.

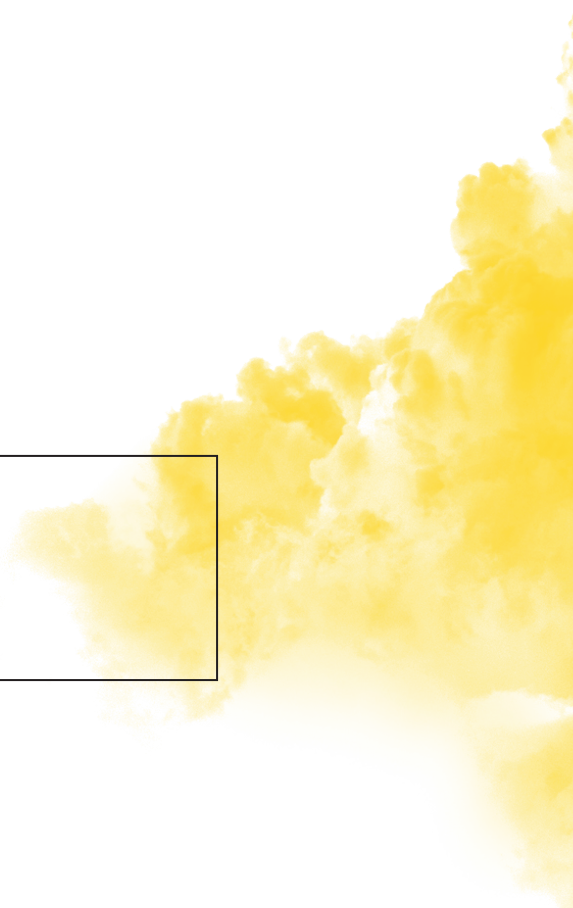
An unrealistic expectation many guys have resulting in frustration and confusion is that testosterone doesn't miraculously transform you into the most masculine expression of yourself once your serum levels reach an optimal level. It takes months, and even years, as well as hard work to reach that potential.

There is a big gap between the expectations and reality of TRT.

I believe many men are looking to remedy this lack of “masculine traits” and think that altering their ratio of androgens to estrogens will fast-track this process. And while it may produce a subjective “benefit” in the short term, it is important to listen to the side effects when they arise, rather than looking to remedy this problem with more pharmaceuticals or supplements.

Never use drugs to fix the side effect of other drugs when the goal is optimal performance, health and longevity.

It is important to also re-iterate this key point: testosterone replacement therapy has been studied long term in men with positive outcomes. Aromatase inhibition, with or without testosterone replacement, has not.

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CHAPTER IV

PREGNENOLONE

4.1 PREGNENOLONE INTRODUCTION

I have spent a lot of time studying pregnenolone, as well as working with people on their pregnenolone supplementation.

This is due to my own personal background with chronic complications from vestibular neuritis and pregnenolone played a crucial role in that recovery.

Since producing two of the most highly viewed YouTube lectures on pregnenolone, training world class clinics in how their doctors and patient care coordinators can treat their patients with pregnenolone, and producing the only comprehensive practitioner training resource, “The Pregnenolone Masterclass”, I have seen how pregnenolone plays out a lot.

The majority of what is discussed and posted around pregnenolone online is inaccurate in the context of hormone replacement therapy.

It may sound great as mechanistic theory in animal or invitro studies, or even in high dose, short term human trials (these trials are very myopic).

But in reality, the majority of people should not supplement pregnenolone, for one key reason.

It is not well tolerated.

While there may be some spectacular, *borderline panacea* effects reported in forums and corners of the internet, there is a reason why widespread use of pregnenolone is not a standard.

In fact, I would wager that the majority of people who have trialed pregnenolone supplementation discontinued it due to side effects.

However, there is a major exception to this.

Pregnenolone deficiency.

When an individual who is truly deficient in pregnenolone restores their levels, the results are nothing short of spectacular.

I can honestly say, I've never seen anything like it.

However, the trainwreck that is the current state of pregnenolone discussions online comes from two incorrect hypotheses.

- ① ***Everyone on TRT needs pregnenolone, as pregnenolone production is reduced when the patient is on TRT.***
- ② ***Pregnenolone's benefits are systemic and everyone can benefit from higher, more "optimal" pregnenolone levels.***

These will be discussed at length in the "myth busting" section, and are the core discussion of the Pregnenolone Masterclass on the Advanced Fundamental Health website.

❓ **So, what is pregnenolone?**

Pregnenolone is often referred to as the "mother" hormone.

It is the hormone from which all other hormones are made, and is the precursor, directly after cholesterol, to form the entire steroidogenic hormonal cascade that contains the majority of the body's hormone production, including cortisol, progesterone, DHEA, testosterone and estrogen.

For a long time, it was thought that pregnenolone only served the role of a precursor and exerted its effect via its metabolism to other hormones.

And while there are no specific pregnenolone receptors in the body, it does exert a variety of neurosteroidal, hormonal, immunomodulatory and other functions in the body via its own mechanisms, both as pregnenolone and pregnenolone-sulfate.

Pregnenolone is often touted online as a “neurosteroid” hormone, due to its powerful modulatory impact on neurotransmission.

It is important to keep in mind that while testosterone and DHT also fit the definition criteria of neurosteroids, pregnenolone and its metabolites not only modulate the sensitivity of the GABA-A receptor, but also play integral roles in the regulation processes relating to dopamine, glutamate, opioid receptors and detoxification processes.

A pregnenolone deficiency often presents primarily as mental illness.

Typically, there are diagnoses of anxiety and depression, as well as insomnia.

Other mental health diagnoses can be made, or behaviours manifest, depending on the person’s personality type and circumstances.

I refer to the low pregnenolone state as “wired, tired and cranked”.

It is very common for these people to experience little joy, peace or satisfaction, as they are constantly overwhelmed, and struggle to function with day-to-day life and career due to severe brain fog. It is also common for these individuals to have been previously placed on a variety of psychiatric medications from their doctors, that will often produce side effects due to the lack of pregnenolone’s protection to neuronal excitotoxicity.

In fact, low pregnenolone individuals will often be hypersensitive to everything, and will often have food intolerances and negative reactions to small amounts of caffeine, as well as alcohol.

This is because they have lost a lot of their biological stress resilience and, as a result, recover poorly from just about everything.

Physical or mental stress, junk food, alcohol - everything hits you hard and leaves you hung out to dry when pregnenolone is low.

The effects of treating a pregnenolone deficiency are life changing.

When it is low, supplementing pregnenolone will create powerful, positive subjective effects on mood, energy, sleep and wellbeing.

I have had clients describe it as life changing, and commonly reported effects include increased perception of beauty such as colour, nature and music, as well as a calmer mind, more positive mental outlook, less anxiety and stress, better exercise tolerance and drastically improved sleep.

Sounds amazing - right?

Here's the catch.

You only get those benefits if you're correcting a deficiency.

? What do you get if you take pregnenolone without a deficiency?

You get the same side effects as someone who is on too high a dose.

Erectile dysfunction, low libido, increased body fat (typically in feminine areas), fatigue, insomnia and reduced positive androgenic effects from the testosterone.

This is because when the body has too much pregnenolone it primarily metabolises into progesterone and creates supraphysiological levels of progesterone.

In men, this is inherently anti-androgenic via reducing 5-alpha reductase (required for the production of DHT) as well as blocking androgens from binding to the androgen receptor.

For most, this is the opposite of what they're trying to achieve with TRT and, it is no surprise that when it is not tolerated, many men report their previous low testosterone symptoms return as a side effect.

4.2 MYTH BUSTING



MYTH #: “Everyone on TRT needs pregnenolone, as pregnenolone production is shut down on TRT.”

If you haven't seen my pregnenolone masterclass yet, I repeat the same phrase upwards of a dozen times in the two-hour session.

“Do not take pregnenolone if you don't have a deficiency”.

If this rule is adhered to 100%, 90% of the problems with pregnenolone supplementation will go away.

(The other 10% is people taking too high a dose)

There is a pervasive theory online, along the lines of: depletion occurs at the end of a cycle of anabolic steroids due to prolonged suppression of LH and FSH. This hypothesis was further elaborated where it was found that the enzyme that converts cholesterol to pregnenolone, CYP11A1, is increased by the activity of LH (becomes zero on TRT or AAS cycles).

This sounds great on paper. However, in practice, it simply doesn't check out.

- 1** There are two main problems with this hypothesis:
If it were true, every man on TRT will be deficient in pregnenolone.
They're not.
- 2** If every man has reduced pregnenolone from TRT and can benefit from restoring its levels, then every man will benefit from at least a small amount of pregnenolone supplementation. They don't.

It is not uncommon for mechanistic theory to not check out in practice, as studies only show a small snapshot of a picture that is a lot bigger with a lot more variables that are not taken into consideration.

In reality, the way pregnenolone works with TRT is very different.

In my opinion, there is likely a small, systemic reduction in total pregnenolone levels that circulate on TRT. This is due to less pregnenolone being created due to the reduction in CYP11A1 via LH's absence. However, as a major end role of steroidogenesis is fulfilled via testosterone (Testosterone, DHT, estrogens).

Less pregnenolone is needed.

And the body is doing this to create balance, as adding more pregnenolone when the body has already regulated its own requirement causes a supraphysiological, unnatural endocrine environment, causing side effects.

I believe this theory relates more to the side effects of prolonged, high dose AAS use, as opposed to therapeutic, testosterone replacement therapy. I wouldn't be surprised if high doses of non-bioidentical anabolic steroids do reduce pregnenolone and DHEA levels long term, but I do not necessarily believe they do either. I believe this was extrapolating n=1 into a widespread hypothesis with the best intentions, but it didn't eventuate.

? So why do SOME guys need pregnenolone on TRT and others do not?

I believe it is because the guys who need pregnenolone on TRT also needed it before they were on TRT and it is part of their overall hormone deficiency.

"Where there's smoke there's fire" and depending on the root cause of deficiency, hypogonadism can occur with, or coincide with, hypothyroidism and deficiencies of growth hormone, pregnenolone or DHEA.

But other times, these hormone deficiencies can occur in isolation.

I have not seen TRT cause a pregnenolone deficiency to date, something I said 3 years ago in my first YouTube lecture and still rings true. I've seen levels drop marginally, but I've also seen them increase or remain the same. I am sure there are exceptions to this rule, but as someone who has worked with a lot of

men on pregnenolone, this is what I have observed.

I have only seen pregnenolone deficiencies in a minority of my clients. I have seen many guys with pregnenolone levels that are not deficient (pregnenolone consistently >50) supplement with doses as low as 5-10mg and not tolerate it due to sexual side effects and lethargy, or anxiety and feminising fat depositions, or emotional side effects.

It is also important to note that LH is not responsible for the primary functions of the adrenal glands, where both pregnenolone and DHEA are produced in high quantities, as well as in other glands and the brain.

There is a study that is further extrapolated and misunderstood, showing that when exogenous testosterone is administered, that production of pregnenolone and DHEA are reduced in the testicles. This is true, as the testicles become inactive on testosterone replacement therapy (unless concurrent use of HCG is used).

However, it is extremely important to identify that this was an isolated testicular cell study, and didn't address systemic levels or functions that could be negatively impacted.

In fact, it is shown that not only does HCG not increase DHEA levels when administered to men on TRT, it has also been shown that TRT monotherapy actually INCREASES levels of systemic DHEA, rather than decreasing it.

When we look at outcomes in practice, monitoring and measuring the levels of these hormones in response to TRT, one thing is clear, there is not a widespread reduction in pregnenolone (or DHEA) levels in response to TRT. If there were, the majority of men would feel lousy on steroid cycles or TRT without taking pregnenolone, and only a small minority take it.

(Remember, as discussed in the HCG chapter, HCG co-administration doesn't significantly increase pregnenolone either.)

To conclude, it is extremely important to note that pregnenolone causes negative side effects if it is not tolerated, which trust me, you want to avoid. I believe there would be more room for debate on this topic if pregnenolone wasn't so poorly tolerated in those who don't need it.

Men lose their libido, gain feminine fat deposition, become emotional, lose the positive effects of testosterone, and develop severe erectile dysfunction and fatigue. The only time these effects DON'T occur and positive effects are gained is when there is an identified deficiency and it is resolved.

There are currently practitioners all over the world who recommend pregnenolone for all patients on TRT and I believe this approach should be avoided. Primarily because I work with guys on identifying the issue, guiding discontinuation and finding they feel better each time. Secondly, because we do not know the long-term effects of elevating these hormone levels to supraphysiological levels for long periods of time.

4.3 PREGNENOLONE VERDICT

Pregnenolone is a powerful tool in the toolbox for any practitioner working in healthcare. However, it requires proper testing, monitoring, titration and supervision, and there is a strong lack of the necessary frameworks in place for it to be used effectively for the most part.

Pregnenolone sulfate, although it is the most accurate pregnenolone test available, is still subject to significant daily fluctuation.

However, an undetectable level almost always predicts a problem.

Repeat low tests, <50, and a history of symptoms that match a deficiency warrant a trial of pregnenolone supplementation.

In my experience, the vast majority of people with undetectable pregnenolone levels benefit from supplementation, on the assumption that dosing is correct.

If someone has a low-end pregnenolone level and no negative symptoms I don't recommend treating it, although I don't think a trial of a low dose for a short period, such as 3-7 days, is a bad idea either. However, I predict that if there are no low pregnenolone symptoms, the person will feel better without pregnenolone, than with pregnenolone added.

If a pregnenolone deficiency is present, it is best treated with oral supplementation in a micronised form. Regular oral works okay, but micronised works far better at improving blood levels and symptoms.

These will ideally be compounded by a pharmacy, and can be put into sustained release capsules that are even better, but by no means needed. A smoother, slower release is already achieved through micronisation, but when it comes to hormone replacement therapy, “fewer peaks and troughs” is always a good approach.

I recommend starting on 50mg of pregnenolone for men (or 25mg for women) if the blood result is undetectable or <50 on more than one occasion. If someone has borderline low levels and wants to trial pregnenolone, I recommend a lower dose of 25mg for men and 10mg for women.

Pregnenolone responses to supplementation vary wildly, and body weight and gender are not the only determining factors. In my experience, most men require a pregnenolone dose of 50-100mg to correct a deficiency, but there is a substantial minority who benefit from 10-25mg and achieve equal blood serum levels. This means that although there are ballparks, pregnenolone dose response varies more than thyroid or testosterone, so more careful emphasis must be placed on testing and follow up monitoring.

I recommend taking pregnenolone at night, as the majority will find it sedating. However, as mentioned previously, pregnenolone supplementation responses vary and a substantial minority find it stimulating and do better taking it in the morning. The effects are often quite pronounced, so it is often easy to work out how it affects you. If unsure, take it before bed.

Pregnenolone can be taken with or without food, and levels can be measured in the blood accurately after a week. Results become more reliable when more exogenous pregnenolone is supplemented, as the replacement dose is far more consistent than the body's natural production.

For those who find it sedating and have a deficiency, pregnenolone will act as a sleep aid.

For those who find it stimulating and have a deficiency, pregnenolone will act as an energy promoting nootropic, like coffee or green tea.

I recommend measuring pregnenolone at either a 12 or 24 hour trough, to make it convenient, but make sure it's clear which one it is and be consistent.

Most will measure their pregnenolone at a 12 hour trough as they have taken their nightly dose. I recommend aiming for a trough level >100. If it is a 24- hour trough, I recommend a trough level >75.

There are many online who will recommend that pregnenolone is optimised to much higher levels, >125-150, or even up to 250. I have only seen side effects when pregnenolone is taken this high due to supraphysiological elevations in progesterone, and pregnenolone's inherently stimulating properties.

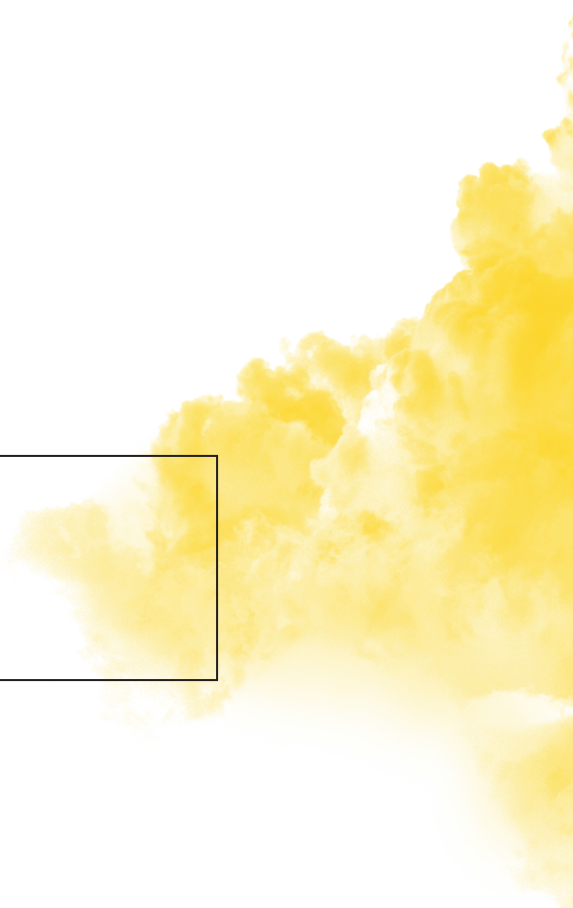
I have consistently seen men with healthy, normal pregnenolone levels of 50-150 develop side effects and not tolerate pregnenolone supplementation at all, even in doses as low as 5-10mg.

This can create a challenge for these men, as they often get both benefits and side effects from increasing their pregnenolone. I hypothesise that this is due to the anti-stress effects of pregnenolone discussed by the late Dr Raymond Peat at great length, and I instead recommend exploring mindfulness meditation, mobility/stretching, as well as supplements such as phosphatidylserine, CBD and KSM-66/Sensoril Ashwagandha. I have lectures on all these supplements on Youtube.

I encourage all men who are considering starting TRT, or who are on TRT and have not previously had their pregnenolone checked, to run a blood test for pregnenolone-sulfate on your next blood panel. It should be fasting tested, first thing in the morning.

I do not recommend taking pregnenolone without the guidance and direction of a practitioner, and I do not recommend buying pregnenolone online and taking it without measuring your levels first.

Although pregnenolone can be measured accurately in the blood after a week, it takes 3-4 weeks for the effects to manifest. I recommend titrating and retesting pregnenolone every 8-12 weeks in increments of 25-50mg (10-25mg for women), and monitoring every 6-12 months once stable levels have been achieved and negative symptoms resolved.

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CHAPTER V

DHEA

5.1 DHEA INTRODUCTION

DHEA falls into a similar category to pregnenolone in a lot of ways.

It's a neurosteroid. It often comes in a capsule. It's usually taken once a day. It can be deficient on or off TRT.

And...

It's often incorrectly prescribed or neglected, leading to undesirable negative symptoms.

I like to describe pregnenolone and DHEA as yin and yang.

Pregnenolone, primarily through its conversion to progesterone and allopregnanolone serves the role of yin, often being sedative, anti-anxiety, and calming.

And while DHEA shares a lot of pregnenolone's anti-stress effects, I put the yang label on it, due to its energising, stimulating and pro-metabolic properties.

It is for these reasons, as well as many others, that DHEA is often sought after in supplement stores (it's over the counter in some countries, but prescription-only in just as many) as, if your natural production of DHEA is low and you supplement with it, generally you "feel" much better, day to day.

DHEA deficiency, in my experience, is more common with old age. In fact, it is uncommon, to encounter a man past the age of 60 who doesn't benefit from at least a small amount of DHEA supplementation.

In fact, this is one thing that sets DHEA apart from the other hormones. It is the one most strongly influenced by age.

In fact, it is quite common for men in their early 20s in the appointments

with me to be concerned that their DHEA is above the reference range and is too high! While in reality, there is no problem, it is just a hallmark of being young, and if anything, it is a positive.

However, this is where some things start to go awry in the internet pseudo-science realm. This is built around the simple, logical fallacy:

“More DHEA = More Benefits” - This is not correct.

(This is discussed at length in the next section of this chapter.)

A commonality DHEA shares with pregnenolone is the incredible benefits of resolving a deficiency.

As DHEA is integral in the production of cellular energy, adrenal function, thyroid function, and the production of serotonin and dopamine while having regulatory properties on the immune system, temperature regulation and opioid receptors, a deficiency leads to a low vitality, depressed and anhedonic state.

Patients or clients with deficiencies often report a cluster of sexual dysfunction symptoms including low drive, inadequate lubrication/erection function, inability to reach orgasm and lack of sensitivity. They will also report low energy, stimulant dependence, low mood and poor stress tolerance.

When rectifying a DHEA deficiency, there can be powerful changes to witness in a client or patient’s physical and mental health. “Wellbeing”, the subjective state of living day-to-day, often improves substantially. Sleep, exercise and energy all improve, and people describe they feel more “youthful”, akin to how they remember feeling in their twenties.

While, in my experience, age is the primary factor for causing DHEA levels to decline, it is not the only factor. There is a substantial minority of younger men I see with low DHEA levels who benefit from supplementation, and reports of many more via other practitioners - it is simply less common.

The causes of these secondary DHEA deficiencies are unknown, but I have witnessed chronic autoimmune conditions, chronic intractable pain, lyme disease, history of cancer treatment and TBI as all being implicated in DHEA deficiency in men of any age.

The downfall of DHEA is quite simple. Practitioners and patients alike look for the benefits of DHEA (mood, energy, wellbeing) without understanding that these benefits don't take place unless they are in the context of restoring a deficiency.

The problem is that taking DHEA when it is not deficient, similarly to pregnenolone, causes side effects. These include acne, increased body odour, insomnia, anxiety, over stimulation, excess sweating, bloating, water retention and other mood disturbances. This is due to DHEA's powerful effects on both androgenic, estrogenic and adrenal functions, and, most importantly the benefits cannot be separated from systemic effects.

5.2 DHEA MYTH BUSTING

"DHEA has x benefit. More DHEA means more of x"

When you look at the benefits of DHEA on paper, it is nothing short of incredible. It would be hard to find an area of health that doesn't benefit from DHEA.

- ❓ Insulin sensitivity? **Yes.**
- ❓ Sex drive and function? **Absolutely.**
- ❓ Mood? **Profoundly.**
- ❓ Immune System? **Yes.**
- ❓ Cancer Protection? **Some say so.**
- ❓ Bone mineral density? **Yes.**
- ❓ Muscle mass and performance? **Yes.**
- ❓ Neurological health? **Absolutely.**

From looking at the benefits of DHEA, it almost seems crazy for someone NOT to supplement with it. So why shouldn't you?

It's very simple, and shares the commonality with pregnenolone.

It's just not tolerated.

It just doesn't work out. If it did, taking DHEA would be as common as taking a multivitamin or fish oil.

There's a reason why most who try DHEA aren't still taking it...

DHEA produces these benefits, and overall systemic improvements in health and wellbeing when a deficiency is corrected. When DHEA is supplemented in a body which is already making healthy levels of DHEA it creates a supraphysiologic DHEA environment. This causes powerful, systemic side effects and leads to discontinuation.

A lot of DHEA's effects occur from improving energy and acting as an antagonist to many catabolic functions of cortisol and adrenaline, so if you are looking for the benefits of DHEA but already have optimal DHEA levels, consider working on diet, lifestyle and stress management.



MYTH #: "TRT reduces DHEA."

In this whole book, this is the best example of a myth, as you will hear it parroted in just about every forum, group and podcast on the topic of TRT.

It is used as a case against TRT. It is used as a case to upsell HCG to "prevent the drop in levels", or it is used to upsell DHEA as a "must have" with any TRT prescription.

In reality? Not true at all.

This is because although DHEA is made in the testicles as a precursor to testosterone, it is primarily made in the adrenal glands. And, as testosterone stimulates the central nervous system, it is common for TRT to (but not always) increase or maintain DHEA levels.

In my experience seeing thousands of blood panels for DHEA, in the majority of cases, TRT will not impact DHEA levels significantly, outside of regular daily fluctuation in DHEA levels.

Does DHEA drop in SOME men on TRT?

Yes. Absolutely. And those men should be treated with DHEA therapy.

However, it is far from common practice, and these men fall into the category of “outliers” rather than the majority.

What is far more common, is men finding out their DHEA is low after starting TRT - however, they did not run a serum blood test prior to initiating TRT to find out that it was already low.

Another scenario is that men will be on TRT for a decade and attribute their TRT to declining DHEA levels, when in reality, it was due to a decade of aging.

MYTH #: “HCG Increases DHEA.”

HCG does not increase DHEA significantly in practice.

This aligns with my observations in working with my clients. To date I have not seen HCG resolve a DHEA deficiency.

This myth comes from the mechanistic theory of the steroidogenic cascade diagram, with the misunderstanding that hormones literally flow down the arrows like rivers, rather than being a diagram for illustrative purposes.

Just as TRT in the majority of men will not significantly decrease DHEA, or impact it much whatsoever, HCG does not significantly increase levels, nor impact levels of DHEA much in practice.

If this myth is reality, we will see low DHEA in the majority of men on TRT monotherapy and optimal DHEA levels in men on TRT + HCG, and this is not an observed hallmark of either treatment.

5.3 DHEA VERDICT

The most common symptom I see of DHEA deficiency is related to sexual function particularly low drive, and low sensitivity that may or may not be involved in difficulty to orgasm.

This is often accompanied by low energy levels, with many similarities in symptoms of subclinical hypothyroidism - low energy, lethargy, always napping and oversleeping, poor stress tolerance.

I don't use the word "wellbeing" very often as it doesn't have an objective, measurable meaning.

However, the best term for what DHEA does for men who correct a deficiency is improve their sense of "wellbeing".

I find DHEA and pregnenolone have similar outcomes in improving overall quality of life, with DHEA helping with the more stimulating and energetic

aspects (yang), while pregnenolone is more involved with rest and relaxation issues (yin).

In fact, the effects of DHEA on dopamine, adrenal function and adenosine share a lot of similarities with caffeine. I often will explain pregnenolone as our body's natural valium, and DHEA as our body's natural, endogenous version of coffee. Being deficient in either one puts you out of balance, while being deficient in both cause you to be wired (low pregnenolone) and tired (low DHEA) at the same time.

It is extremely important to check for both pregnenolone and DHEA deficiencies as they equally often occur together as they do in isolation. And it is extremely important to understand that the client or patient's negative side effects will not resolve unless all hormone deficiencies are resolved.

In my experience, DHEA levels are best, as a rule of thumb, around mid-range.

In my experience, DHEA is best left alone if levels are not low, and while bringing levels from low to middle confers a strong benefit, increasing DHEA from moderate to high causes problems, and taking DHEA from low to too high causes the same problems. These issues are typically agitation, insomnia, bloating, mood disturbances and acne.

As DHEA is an adrenal hormone and is especially related to the functions of the adrenal glands, to me it makes sense why it has this "goldilocks zone" characteristic. Similar logic applies to other adrenal hormones such as cortisol.

DHEA can be taken as an injectable, oral, sublingual or transdermal supplement. While it can be purchased online in most countries, I always

recommend a compounded pharmaceutical preparation where possible. If a commercial preparation is used, I recommend a micronised version as it has a more gradual release. I recommend the brands MRM and Nutricology.

My preference for DHEA is micronised, oral supplementation taken in the morning.

There may be some fringe benefits to topical or sublingual, but I do not believe these benefits are worth the inconvenience of not having a long term, straightforward solution. DHEA can be taken with food or on an empty stomach.

DHEA peaks naturally in the morning, and as it inherently has many activating properties, I recommend a once daily AM dose. I have seen many recommend taking DHEA at night as one study shows it increases growth hormone when taken in this fashion. However, in practice, DHEA will worsen sleep quality in the majority of men who take it at night., which is the simple reason why it should be taken in the morning.

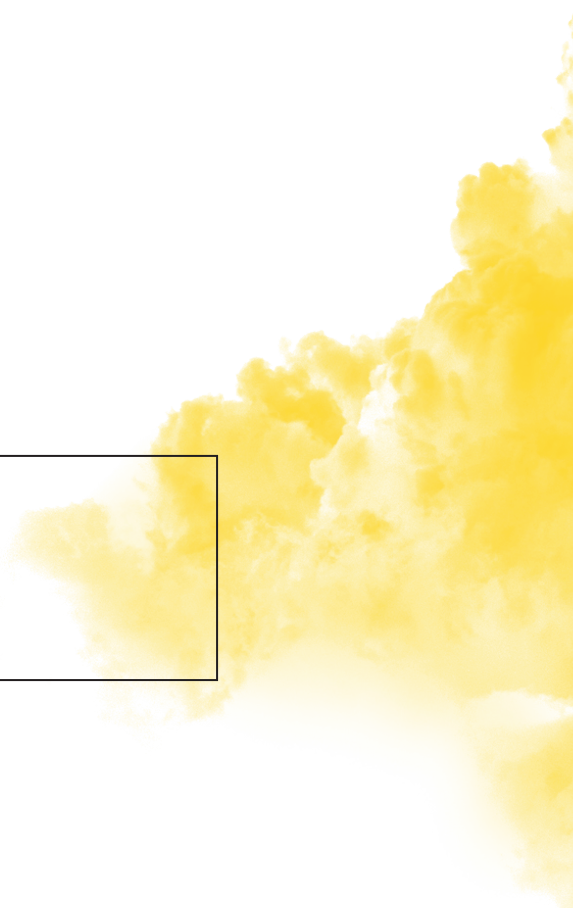
I find DHEA is best optimised to levels of 4-8 nmol/L. In my experience, taking levels to above 8nmol/L (at 24-hour trough) causes side effects, as levels are peaking much higher when taking the daily dose.

When measuring at a 2-4-hour peak, I find DHEA is best optimised at 6-15. This is a broader range, however, due to the unstable pharmacodynamics of oral, non esterised hormone supplementation, I prefer to measure and optimise DHEA based on a 24-hour trough, rather than peak.

I recommend initiating DHEA supplementation at 10-25mg for men at 5-10mg for women. I recommend titrating DHEA dosing by 10-25mg increments. I find that most men require 25-50mg to achieve optimal

DHEA levels when they are deficient while using high quality, pharmaceutical preparations. There are a decent minority of men who will achieve optimal levels on 5-10mg, as well as some men needing up to 100-150mg to achieve optimal levels.

DHEA can be felt instantly for many men, or with others noticing the benefits building over a 2-4-week period. DHEA-s (DHEA Sulfate) is a 24 half-life metabolite of DHEA which is common practice to measure in the blood, and can be accurately measured after as little as 1 week of supplementation. However, as with pregnenolone, it is crucial to titrate in 2-3-month increments, as it can take many weeks for the subjective benefits of DHEA to be fully realised.

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CHAPTER VI

FERTILITY

6.1 FERTILITY INTRODUCTION

Male fertility is an under-discussed topic outside of HRT, and more men than ever before are finding issues with conception due to complications from lifestyle, diet and endocrine factors.

In some cases, men will even discover their hypogonadism while investigating their poor fertility parameters. As I have written previously in this book, “where there’s smoke there’s fire”.

Unfortunately, while this concept goes both ways, perspective is often lost, and in recent years, the discourse around the topic of male hormone replacement therapy and fertility has changed to become quite alarmist and misleading; **Many men on TRT have fertility issues. However, many men don’t.**

Here’s the important part:

Many of the men on TRT with fertility issues had undiagnosed, uninvestigated fertility issues before starting TRT. And it often has the same root cause as the diagnosed hypogonadism - poor diet and lifestyle, obesity, drug use (including pharmaceuticals) and, of course, age.

It is also important to address another key point - the men who don’t have fertility issues are not the those talking about it in the forums you land on after google searching the topic.

Most people who discuss HRT online aren’t sharing success stories; they’re trouble shooting.

If testosterone replacement therapy was a reliable method of male birth control, it would be MUCH more popular, and applied for this specific purpose.

In fact, there are many body builders who have assumed that due to testicular

atrophy or diminished ejaculate volume while on a supraphysiological steroid cycle, that they are infertile. These men often became fathers as a result.

While testosterone replacement therapy can complicate fertility, or in some instances make a man temporarily infertile, it does not do this in all cases. And this is what leads to much of the confusion regarding the “will it, won’t it?” question of infertility and TRT.

6.2 FERTILITY MYTH BUSTING



MYTH #: “TRT makes all men infertile unless you use HCG.”

This is the biggest myth of all, and I believe this is primarily promoted by clinics that are selling HCG as an upsell for all their TRT programs.

The best marketing strategy is to create a problem you didn’t know you had and sell you the solution.

When I started TRT in 2016, the discourse was very different, and much closer to the truth, as discussed in Jay Campbell TOT Bible.

TRT may certainly cause fertility problems, but it is far from a guarantee.

This is supported by the many men who conceive healthy children on TRT without any form of fertility support. We just don’t hear about these men as often.

As mentioned previously - if this is a hard fact, we will be using TRT as a method of male birth control and it would likely be VERY popular for men of all ages.

The second part to this myth that needs busting is that HCG as a TRT ancillary is not a bulletproof vest against infertility. So, while running HCG with your TRT might make you “feel” better, in reality, I have not seen this to consistently preserve fertility.

Similarly, I have not seen TRT monotherapy consistently cause infertility.

This is because HCG primarily activates the LH receptor, that only plays a supportive role in facilitating spermatogenesis (factors relating to sperm production and health). HCG has some mild, spillover effects into the FSH receptor at the doses we typically use for intratesticular testosterone production, this is why the doses used of HCG for TRT (typically 500-1500IU a week in divided doses), we only receive a mild impact on this pathway.

In my experience over the last three years, I have seen many men on TRT who are infertile.

? *Have I seen more infertile men on TRT monotherapy than TRT + HCG?*

No. It's roughly the same.

In fact, one of the surprising observations I had early in my career as a health coach was that the mechanistic theory of HCG preserving fertility as a TRT adjunct in reality was more of a "it may help a bit", rather than a "sure thing".

This is a stark contrast to how HCG is marketed. So, understand this:

TRT has a significant change of complicating fertility. It is not guaranteed to render you infertile, but it does have a good chance of reducing fertility to some degree. If you become infertile as a result, your pre-existing fertility status was also a factor, but not the only determinant. Individual variability, biochemical individuality, n=1, or "luck", seem to all be factors in the outcome.

I recommend that men who want to have children in the future consider getting a fertility test before starting TRT, as well as 6 months into treatment, and discuss and explore options for future fertility protocols depending on the timeline that they're looking to have children.

For most circumstances, as a general rule of thumb, I recommend delaying TRT until after successful conception if you are planning children in the next 12 months. Don't potentially complicate an already complicated process unless you have to.

If children are years away, consider using the time in the interim to correct your hormonal issues and get your diet and lifestyle squared away to build yourself into the best future-father you can be, and then address fertility when the time comes.



MYTH #: "If you take HCG every "x" interval, you'll be more likely to conceive in the future."

This segues into the next myth.

A very common scenario for me to see with men who are concerned around fertility on TRT, is finding out they're infertile and wanting to conceive in 2-5 years, or beyond.

This is a difficult circumstance, because depending on who you ask, will depend on the answer you get about what to do.

Especially if the individual reacts badly to HCG.

I am a big believer in the notion of "test not guess" in these scenarios.

For example, if you are utilising HCG for remaining fertile while on TRT - ***test your fertility!***

Some practitioners will suggest that HCG is used cyclically, in various approaches such as once a month, once a quarter or a 2-3 times per year, to prevent the testicles from remaining "dormant" for a prolonged period of time, that will potentially lead to better fertility parameters in the future.

Unfortunately, others nor I, has a clear-cut answer on the validity of this approach. It has not been studied.

My concern is that if the individual reacts badly to HCG, that even short use periods could be causing side effects that may negatively affect the individual in the future. Again - this has not been studied.

My comment on this area is that in practice, I have not seen a difference in the success of the acute fertility protocols (discussed later in this chapter) based on whether the individual has used HCG periodically or not.

I have found these fertility protocols to be successful regardless, with the only factor being their pre-TRT fertility status.

I believe there is merit to the theory of this approach, but for now, this myth will remain non-definitively “busted”, but in my anecdotal personal experience, although it is *theoretically* sound, I have not found it to be necessary.



MYTH #: “You have to come off TRT to conceive.”

Visit 99% of fertility specialists, urologists or endocrinologists and they’ll give you this same line.

Want to fix your fertility? Stop your TRT cold turkey and it’ll come back in 6-12 months.

I’ve seen countless guys go through this and the outcomes on their physical and mental health were devastating. I also have concerns that they’re conceiving their child in an unhealthy, hypogonadal state that may impact the epigenetics of the child.

You want to be at your best when you conceive, not at your worst.

However, does this solution work in restoring fertility? Absolutely it does.

There’s just a lot of unnecessary suffering to go with it, and men are not being informed that there are other options that don’t involve them ceasing their TRT.

And for men with a history of poor mental health, this is especially important, as this crash of testosterone, combined with the added pressure of infertility, can be psychologically devastating for many men.

Instead, I believe all men should consider the following treatment option, before assessing how to move forward with their fertility concerns.



MYTH #: "High dose HCG is the best way to improve fertility."

Wrong. FSH + HCG is the best way, at the time of writing this book, to improve fertility.

I believe that if bioidentical, (ideally esterised) LH becomes readily accessible, the combination of exogenous LH and FSH will be the future gold standard of fertility protocol for TRT.

One way to increase fertility on TRT is to use higher doses of HCG than are required to maximally stimulate LH receptors (250IU per shot), and leverage the spillover effect into the FSH receptors.

I have seen 500IU per day be common practice for this approach, however, the studied dose is 1500IU 3x weekly, and I have seen much higher doses than this used in an attempt to regain fertility.

The primary issue with this is that many men do not tolerate large doses of HCG well.

As discussed earlier in this book in the HCG chapter, I believe this is due to large doses of female pregnancy hormone negatively reacting with male biology, which is especially noticeable when that man is seeking to improve his androgenic nature and traits.

The much better, bioidentical and side-effect free option is bioidentical FSH. It is literally replacing the primary signal operating fertility, that shuts down while on exogenous testosterone.

Rather than crashing a man's testosterone levels to regain the natural levels of this hormone, it can simply be replaced and optimised for a short period while the man is trying to conceive, and then withdrawn after conception while he retains optimal testosterone levels through his TRT.

In my experience, using an effective dose of FSH (50-150IU 3x weekly) in conjunction with 250IU daily of HCG optimises testicle function for both intratesticular testosterone production and spermatogenesis, and restores a man to his pre-TRT fertility levels.

And as mentioned previously, I have found periodic or long-term low dose HCG use has not been a determining factor as to whether the outcome of this protocol is successful.

 ***The determining question is - was he fertile before TRT?***

6.3 FERTILITY VERDICT

Fertility is a sensitive topic for many men, just like hypogonadism.

Unfortunately, due to fertility's integral role in many men's life purpose and starting a family, it can be an area where irrationality in behaviour and thinking can be ignited by fear.

I encourage all men to obtain a fertility test BEFORE and AFTER starting TRT prior to addressing any discussions around fertility with their provider. And, to reiterate, if conception is on the horizon in the near future, simply delay starting TRT until after you conceive.

Fertility is an extremely complex process and involves optimal health parameters for both parties. So, don't forget to measure your partner's fertility when you are looking to conceive, as if there is a problem on the woman's side, it will not matter how much HCG or FSH you throw at the problem on your end.

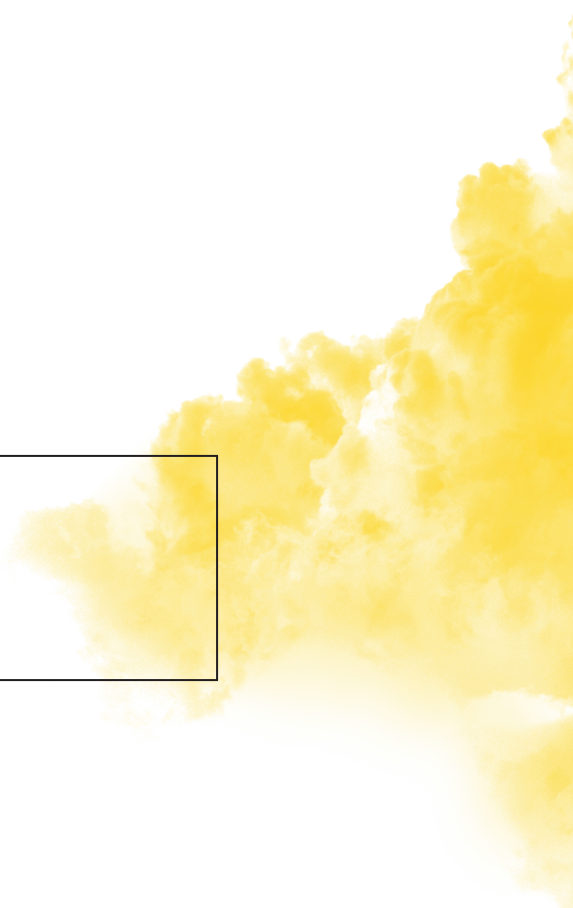
FSH has a minimal effective dose of 50IU 3x per week, and the most commonly studied dose and the point of diminishing returns is 150IU per week. Depending on your budget and the cost of FSH (it often doesn't come cheaply) you may be better off starting with 50IU per week, and retesting your fertility parameters after 12 weeks, to see if you need to increase to 100-150IU 3x weekly to achieve optimal parameters. In my experience, the minimal effective dose of 50IU 3x weekly can be sufficient in some cases, but many will need the full 150IU to achieve optimal fertility. However, it does not always take a fully optimal sperm count to conceive.

I recommend combining this with 250IU daily of HCG, or if HCG is poorly tolerated, 25-100IU of HCG per day.

I recommend that if men are planning to conceive, to come off all unnecessary pharmaceutical medications and anabolic steroids, as well as recreational drugs and alcohol, and get yourself into the best physical and mental shape as possible before conceiving, to ensure you have the healthiest baby possible. The health of the mother and father is being found to have more of an impact on epigenetic expression the more it is researched.

If FSH is not affordable, it is important to consider whether starting a family is affordable.

Remember: Stress, poor diet and lifestyle, drugs, alcohol, poor sleep and obesity will all negatively impact your fertility, even in the presence of fertility protocols.

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CHAPTER VII

THYROID

7.1 THYROID INTRODUCTION

The thyroid is a system that is commonly overlooked and poorly understood, especially in men.

When assessing the list of symptoms of hypothyroidism, it can often mirror the lists of symptoms for conditions including, but not limited to, major depressive disorder (especially anhedonic depression), generalised anxiety disorder, chronic fatigue syndrome, “adrenal fatigue”, inattentive ADHD and metabolic syndrome.

? *Have you ever met someone who says "I've just got a slow metabolism"? That's called undiagnosed hypothyroidism.*

And it's a common problem.

This is for three primary reasons:

- 1** There are many diet and lifestyle factors causing secondary hypothyroidism, that creates a vicious cycle of mental and physical symptoms.
- 2** Autoimmune hypothyroidism can be genetic and triggered by environmental factors, stress or unknown causes, and is often hereditary.
- 3** Doctors do not properly test for, or treat, subclinical hypothyroidism.

Even if a patient is lucky enough to get their thyroid properly tested and receive treatment, it is more common than not to receive T4 monotherapy.

This is why thyroid treatment has such a bad reputation online. Patients take their T4 monotherapy as prescribed and it's either too weak and does nothing, or it's too strong and causes more harm than good.

The thyroid, especially T3 (the active form of thyroid hormone) is the main governor of your body's energy metabolism. A low energy body and low energy brain is, by definition, a depressed body and brain.

“Stress” is defined by not having enough x to do y. This can be not enough money to pay the bills, time to get things done, or in the body’s case, not enough cellular energy to perform the biological and physiological functions required.

And this situation leads to, and is worsened by, excess body fat.

If the engine is too small for the car, a bigger car is going to make the situation worse.

Symptoms of low thyroid can be felt in the body and the mind. As thyroid hormone is a major lever in dopamine metabolism and neurological energy metabolism, it is very common for hypothyroidism to manifest psychologically and impact mood and behaviour.

Many patients with undiagnosed and untreated (or undertreated) hypothyroidism struggle with focus and attention, as well as performing other tasks related to the pre-frontal cortex such as impulse control, emotional regulation and attention to detail. When we are in a low energy state the body will often upregulate adrenaline function to compensate, leading to a hair-trigger fight or flight response, that can manifest as a state of chronic, generalised anxiety.

This wired, tired, overwhelmed and low energy state can lead to avoidant behaviours, procrastination, as well as guilt and low self-esteem.

While this state ticks almost every box for “inattentive ADHD” (I truly believe this diagnosis can be resolved by diet, lifestyle, proper endocrinology and discipline) and while hypothyroidism does respond well initially to ADHD stimulant medications such as amphetamine and methylphenidate in the short term, this does not resolve the root cause issue and is prone to well-known short term and unknown long term side effects.

If you have been diagnosed with inattentive ADHD, I thoroughly recommend running a comprehensive hormone evaluation.

While this state ticks almost every box for “inattentive ADHD” (I truly believe this diagnosis can be resolved by diet, lifestyle, proper endocrinology and discipline) and while hypothyroidism does respond well initially to ADHD stimulant medications such as amphetamine and methylphenidate in the short term, this does not resolve the root cause issue and is prone to well-known short term and unknown long term side effects.

If you have been diagnosed with inattentive ADHD, I thoroughly recommend running a comprehensive hormone evaluation.

However, hypothyroid symptoms are often physical in their primary presentation.

Individuals with hypothyroidism have a lower basal metabolic rate. This means they burn fewer calories than they should at rest, as well as when they exercise.

? ***How does this feel?*** *Exercise feels extremely difficult and more stressful than it should, and no matter what diet and macro protocol you follow, you can't lose weight unless you crash diet.*

These are the individuals who give up on diet and exercise protocols, and personal trainers often chastise them for not “tracking their sauces” properly in MyFitnessPal as the reason they're not losing weight on their plan.

Hypothyroid individuals often self-medicate with reliance on stimulants that solves one problem but creates two more. Or, they resort to naps and a low-energy-output lifestyle (lazy) that can lead to poor relationships, career and self-fulfillment outcomes, as well as depression, shame, anxiety and low self-esteem.

Hypothyroid symptoms and TRT look very similar. In fact, almost every symptom of hypogonadism and hypothyroidism overlap. This is crucial to understand.

? *Why? Because the “cut-off” for what is, or isn’t low testosterone is based on blood work AND symptoms.*

This means that borderline low testosterone with hypothyroidism can look like hypogonadism, if the thyroid is not assessed.

This must be understood and applied to the practice of TRT, as untreated hypothyroidism will cause low testosterone.

This means that secondary hypogonadism induced by hypothyroidism can and should be resolved (or at least, attempted to resolve) before committing to testosterone replacement therapy.

Many patients start TRT only receive partial improvement in negative symptoms, and receive the “full benefits” of testosterone when their thyroid is also improved.

I liken this to thyroid “putting the wind in the sails” of TRT.

Secondary hypothyroidism is subclinical in nature (not severe, but still enough to cause symptoms) and can also be induced by the same root cause as low testosterone. Environmental toxins, poor diet and lifestyle habits, alcoholism, pharmaceutical medications and undiagnosed, untreated chronic disease states can all slow down thyroid production as part of their pathologies.

As can chronic stress.

In my health coaching practice, I have found chronic stress, whether it is physiological (chronic pain and injury inflammation, prolonged malnutrition) or psychological stress to be the most common cause of subclinical hypothyroidism.

This will often manifest with elevated TSH above the “functional range” (>2.5), high T4 and T3, with symptoms of hypothyroidism. Follow up testing often confirms elevations in prolactin, cortisol and most

importantly, Reverse T3, part of the body's stress response to slow the energy metabolism and conserve resources.

While throwing more T3 at the situation can help with some symptoms, it does not work on the root cause, and resolving the root cause of stress, or at least supporting the body as much as possible, is the best way to reduce RT3, and thereby increase T3 activity at the receptor, and reducing TSH as a result.

In these cases, I recommend supplements such as magnesium, CBD, Phosphatidylserine and KSM-66 Ashwagandha, in conjunction with daily stretching, mobility and mindfulness meditation, while working on resolving the root cause of the stress.

7.2 THYROID MYTH BUSTING



MYTH #: "Synthetic T3/T4 is the same as prescription natural dessicated thyroid (NDT)"

While synthetic, bioidentical T3 and T4 are perfectly sufficient thyroid therapy options, in my experience, it is outperformed by NDT.

There are theories as to why this may be the case, surrounding the other present thyroid hormones active in NDT, peptides, enzymes and RNA in the NDT and increased bioavailability due to presence of cofactors.

One "grain" of NDT is 60mg and contains 38mcg of T4 and 9mcg of T3. Personally, I round this up to 40mcg of T4 and 10mcg of T3 for ease of explanation and calculations when I am explaining dosing.



Quick fun fact: It's called a grain because 60mg is about the same weight as the average grain of wheat. Great party conversation.

In my experience, I have found that NDT can be ~50% more effective than its equivalent T3/T4 dose.

I have also found Hashimoto's patients respond better, both in terms of lab improvements and symptom resolution, to NDT, and when using synthetic preparations often have to push their T3 >50mcg and T4 >200mcg to get results, while often achieving better lab work and symptoms on 4 grains of NDT, or sometimes even less.



MYTH #: "You must take your thyroid dose on an empty stomach twice daily."

Yes to the first part.

To the second, no.

Firstly, thyroid **MUST** be taken on an empty stomach.

Many nutrients, but especially calcium, will disrupt thyroid absorption, so thyroid must be taken fasted, at least 30-60 minutes before consuming other supplements, coffee or food in the morning.

I recommend you put it on your bedside table and take it when the alarm goes off.

This complicates afternoon dosing in terms of absorption and compliance, and I propose the argument that as the half- life of T3 is still 24 hours, that split dosing does not give a net positive in terms of benefits vs compliance.

i.e., "the juice is not worth the squeeze"

Thyroid therapy works perfectly well with single morning dosing for most patients, especially milder cases of hypothyroidism, and is much easier for long term adherence.

Many times, I have seen split dosing recommended to try and fix symptoms, while in reality the patient simply needed a higher dose of thyroid hormone, or needed to address other issues.



MYTH #: "Waking temperatures should be between X and Y."

Waking temperatures are a useful tool to help assist with determining thyroid function.

However, many things outside thyroid function will impact waking temperature, and a low waking temperature does not automatically mean hypothyroidism.

In my experience however, an *optimal* waking temperature does often rule out hypothyroidism.

This is why it is important for patients to not self-diagnose and self-medicate, and for practitioners to gather as many data points as possible in order to create a meaningful diagnosis and treatment plan for the patient that will deliver the optimal results.

The optimal waking temperatures are believed to be 97.8-98.2 °F (36.5-36.7 °C), with some variance depending on who you ask.

I have found that many individuals will have optimal thyroid function (and therefore do not tolerate additional thyroid hormone), even if their waking temperatures are in the 97s °F (36s °C), as their low body temperature can be caused by other factors, especially digestive problems.

My main area of clinical recommendations for waking temperatures is when fine tuning the thyroid dose, ensure the dose has not overshoot the mark, and that the waking temperatures are not too high.



MYTH #: "Thyroid antibodies that are elevated, but not above the reference range do not require treatment."



And if someone is 1% lower than the cut off for diabetes, they're perfectly healthy too right? Wrong.

Hashimoto's Thyroiditis is an autoimmune condition where the body attacks the thyroid gland. It is more common in women, and it very commonly occurs in conjunction with coeliac disease.

But it does occur in men and is why it is always vital to screen for thyroid antibodies (and reverse T3, a full thyroid panel) when wanting to diagnose and understand thyroid function.

If Hashimoto's is caught early before symptoms become debilitating, the antibodies may be within range, but still elevated.

This should not disqualify someone from treatment for hypothyroidism, this should be a marker to LOOK FOR symptoms of hypothyroidism to maintain an optimal thyroid state to prevent unnecessary manifestation of the diseased state.

The symptoms will be milder and the TSH/T4/T3 levels may not be as high/low/low as someone who has severe, untreated Hashimoto's, but early treatment prevents the unnecessary damage to the body, mind and life of the individual by keeping the car on the road rather than waiting for it to crash.

Often, these men benefit from doses such as 15mg, 30mg or 60mg of NDT along with supplements like KSM 66 Ashwagandha and Phosphatidylserine, as well as a high protein diet.

7.3 THYROID VERDICT

If I had a dollar for every time I found someone who was trying to fix their undiagnosed hypothyroidism by cranking up their TRT dose...

Thyroid treatment is far less invasive than TRT, as it is an oral, daily capsule/tablet, where natural production is not fully suppressed and will restore upon cessation.

As it is less invasive, more straightforward and easier to access, thyroid should always be assessed when a patient presents with low testosterone symptoms as part of a comprehensive screening to identify exactly where the symptoms are coming from before initiating TRT.

Testosterone is much more marketable, and what a lot of men want, but in reality, thyroid may be what they need.

Or, both.

I recommend initiating thyroid therapy at 30-60mg of NDT and titrating up in 30-60mg increments to achieve symptom resolution, an optimal T3 level (within a point either side of the top of the reference range as a ballpark example) and a low TSH (1.5 or less, but not undetectable). This is a “rule of thumb” and an ideal scenario.

If T3 or T4 is used, I recommend an introductory dose of 50mcg of T4 and 10-12.5mcg or T3 (standard preparations are 50mcg or 100mcg T4 tablets and 20mcg or 25mcg T3 tablets) and working up in similar increments.



***Note:** TSH may not receive feedback properly from T3 due to pituitary issues or Hashimoto's, and in more complex/advanced pathologies, it may require complete suppression to optimise T3 levels and resolve negative symptoms. This can be advantageous in Hashimoto's, as Hashimoto's may produce volatile thyroid hormone output in some patients. This is why it's important to listen to the patient's symptoms, and when in doubt, check the waking temperatures, Reverse T3 and antibodies.*

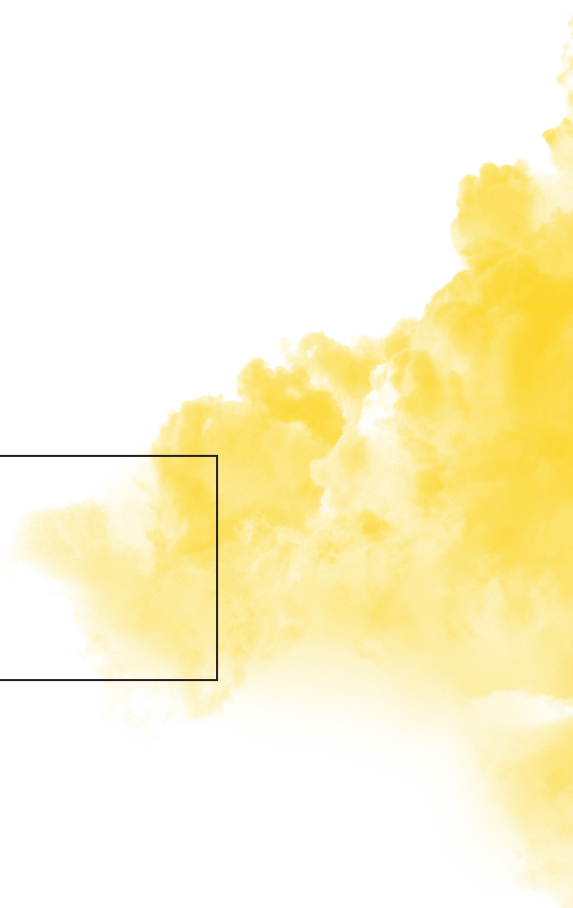
When Reverse T3 is elevated as part of a chronic stress scenario, I recommend the supplements EPA/DHA combined 2-3g from high quality, concentrated Omega 3 fish oil, in the morning, Phosphatidylserine 200-600mg before bed, KSM 66 Ashwagandha 300mg twice daily, and 10-50mg of CBD before bed. It is also integral to focus on resolving the root cause of the stress, such as mobility work, physical rehabilitation, sleep hygiene, resolving insomnia/sleep apnea, or addressing personal/career issues.

In my experience, subclinical, non-autoimmune hypothyroidism can often be resolved with 1-2 grains of NDT or 50-100mcg of T4 and 10-25mcg of T3. However, Hashimoto's often requires a higher dose, up to 4 grains of NDT and occasionally even more.

Thyroid can be tested every 12-16 weeks and should only be titrated and adjusted in periodic reviews with supervision.

I prefer to measure thyroid hormone labs at peak, 2-4 hours after taking the medication, but I will also measure at a 24-hour trough if I am troubleshooting, as some individuals (uncommon) will metabolise the medication quickly and must split their dose in the morning and afternoon.

Over the counter NDT preparations that contain raw desiccated thyroid, often in a 30mg dose vary in concentration of thyroid hormone per capsule, and are not suitable for clinical presentations of hypothyroidism long term, but can be a good place to start as a band-aid while getting access to a proper prescription and supervision. I recommend introducing these at 30-60mg per day, and titrating up to 120mg before moving over to a prescription-based product long term. For those wanting to boost their metabolism temporarily to further body composition improvements in a caloric deficit, this would be a better option than prescription thyroid hormone, if there is no clinical deficiency.

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CHAPTER VII

CONCLUSION

CONCLUSION

There's a big difference between TRT, and TRT *done right*.

For a substantial number of guys, a simple, cookie cutter TRT monotherapy protocol will be just fine. Even a once-a-week injection, while suboptimal, will be "ok" for many guys on TRT.

The current issue is twofold:

- 1 Clinics and providers overcomplicating the process, whether to sell add-ons to bump their profit margin, or based on a gross misunderstanding of endocrinology.

This results in patients and clients taking three, four or five unnecessary medications causing side effects and complications, and leading them to being worse off than before they started.

- 2 On the contrary, there are guys out there who need more elaborate, complex and nuanced hormone replacement therapy due to complicating factors not present in the average low testosterone man.

- 3 *So, are your negative symptoms psychosomatic? Are they coming from your low pregnenolone, DHEA, sub optimal thyroid? Or are they coming from your diet and lifestyle, or even a combination of all of the above?*

You don't know. And you're not supposed to.

This e-book is intended to provide a 101 overview of these complex topics to give you a framework and guide for troubleshooting these issues and avoiding the pitfalls that plague many men in their quest for getting their health to an optimal place.

If it hasn't been made clear yet - please, have your TRT diagnosed and managed by a professional.

At the time of writing this book, these are my recommendations for the best providers for men's HRT worldwide:

AUSTRALIA

Primal Zone

UK

Eden Clinic

Felix Hay

Balance My Hormones

AMERICA

Marek Health

The Restore Clinic

Elevate Men's Clinic

Dr Jeffrey Rutersbusch

Dr Jordan Grant

INTERNATIONAL RESOURCES

Jay Campbell: Podcast + The TOT Bible

The TRT and Hormone Optimisation Channel Youtube

Ali Gilbert

The Silverback Summit

Atlas Optimization

I hope that this book has armed you with the knowledge and basic understanding of these topics to help guide your health journey, and to avoid the many potential pitfalls that may befall you.

Remember:

Test, don't guess (*Get blood work and see an informed provider.*)

When in doubt, take it out (*Don't take any medication or supplement without discussing side effects with your provider.*)

And finally,

Take care of yourself.

Thanks for reading.

Dave

A handwritten signature in blue ink, appearing to be 'Dave', with a stylized, flowing script.

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www.advancedfundamentalhealth.com

"There are only a few times I can remember where I've felt like a health professional has actually genuinely cared about my long-term health and wellbeing beyond just being another patient, and only a single time that they've been really committed to diving to the root of all my problems and seeing the big picture instead of band aiding all my symptoms and calling it a job well done. All you need is one conversation with Dave or a week looking at the work he does in the health space to know that his absolute passion is helping others and making them the best versions of themselves. Dave listens, asks questions, gets extensive blood work, examines all possibilities and dives to the root of the problem, fixing it with a holistic approach. He pointed out many things in my health that no other health professional even considered could be an issue. Dave has already made an enormous change in my health in less than 5 months and things just keep getting better. Where more conventional pathways have failed me, Dave has turned my life around and set me on a trajectory where I can finally be happy and healthy."

— Jonathan M

"I have worked with Dave for a few years now. Not only have we dialed in and drastically improved my health, he has helped me to become a better man in physical, mental and spiritual aspects. Truly grateful to be a client of Dave's, a man who truly cares and is results driven."

— Luke S

"Sending lots of love and appreciation out to the one and only Dave Lee. Just wanting to say a big thankyou to you Dave. For the ongoing work you do for the group of guys in this group, beyond this group, and in particular for what you've done for my personal life. Since first consulting with you I feel like my search ended and this is now where I'm at:

Psychological health: best it's ever been. Massively reduced social anxiety. Improved general well-being and self-belief.

Emotional health: best it's ever been. Now experience overwhelming feelings of happiness every now and then sometimes to the point of tears. Generally, I say to myself I am happy.

Physical health: Best physique of my life. Most flexible I have ever been in adult life. Cardiovascular health feels strong I can jog for an hour easily / surf / play with my kids. Gut health: I had digestion issues which have now gone.

Business: best position I've been in my business since starting it in 2008.

Dave you've not influenced my personal life in a huge way but I also pass on any relevant information to close family and friends. There is a wave like influence out from what you do.

How has all the above happened?
All I've done is do whatever Dave has told me to do when he's told me to do it. My first consult was in December 2021. I truly believe if everyone's local GP was more like Dave then the world would be a far better place.

Thanks Dave from the bottom of my heart."

— Ian G

Dave was referred to me by a mutual friend of mine regarding questions around post cycle therapy. Dave called me directly and actually offered free advice and ended the call. I was shocked because I was expecting him to charge me but this guy with his wealth of knowledge just helped a total stranger. After that experience I ended up booking with him to learn and gain more. He is a wealth of knowledge and am very grateful for what he's taught me.

— Jonathan L



PREGNENOLONE, DHEA, THYROID,
ESTROGEN, HCG & FERTILITY



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